

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA
VIZIANAGARAM

College of Engineering Vizianagaram (JNTU-GV CEV)

Dwarapudi, Vizianagaram – 535003, Andhra Pradesh, India | Official Web Portal: <https://jntugvcev.edu.in>

DOCUMENT 1 — TECHNICAL SYSTEM HANDOVER & MAINTENANCE
DOCUMENTATION

Comprehensive Architecture, Technology Stack, 99 Database Tables, 85+ Routes, RAG AI Engine & Maintenance Manual

JNTU-GV COLLEGE OF ENGINEERING
VIZIANAGARAM

TECHNICAL SYSTEM HANDOVER & MAINTENANCE DOCUMENTATION

Project Official URL: jntugvcev.edu.in | System Version: 2.0 (Production Release)

Classification: Institutional Technical Documentation (Developer & IT Team Manual)

1. DOCUMENT CONTROL & METADATA

Attribute	Specification Details
Project Name	JNTU-GV CEV Official Institutional Web Portal & CMS Redesign
Application Name	JNTU-GV CEV Web Platform & AI Campus Assistant
Institution / College	Jawaharlal Nehru Technological University Gurajada Vizianagaram — College of Engineering Vizianagaram (JNTU-GV CEV)
Official Domain / URL	https://jntugvcev.edu.in (Staging IP: http://89.116.134.182)
Software Version	Version 2.0 (SSR + Dynamic Block CMS + Vector RAG Integration)
Handover Date	August 2026
Project Guidance & Direction	Sri / Dr. W. Anil Sir (Assistant Professor / Faculty Coordinator)
Development Team	- Likhith Kumar Mankala (Developer) - Ch Sai Rupini (Developer) - P Anitha (Developer) - P Sai Vamsi (Developer)
Technical Support Contact	support@jntugvcev.edu.in / principal@jntugvcev.edu.in
Repository URL	https://github.com/Likhith32/JNTU-REDESIGN.git
Document Classification	Strictly Confidential — Technical Handover (Zero Financial Data)

Document Revision History

Version	Date	Authors	Primary Changes & Milestones
1.0-alpha	Nov 2025	Dev Team	Initial React 19 + TanStack Router migration & static UI prototypes.

Version	Date	Authors	Primary Changes & Milestones
1.5-beta	Mar 2026	Dev Team	PostgreSQL integration via Drizzle ORM, 99 database schemas, <code>/admin</code> portal.
2.0-rc	Jun 2026	Dev Team	HuggingFace vector embedding RAG engine (<code>rag_chunks</code>), dynamic Department Block Builder.
2.0-final	Aug 2026	Dev Team	Production release, full SSR hardening, security audit, and formal institutional handover.

2. PROJECT OVERVIEW & SCOPE

2.1 Purpose of the System

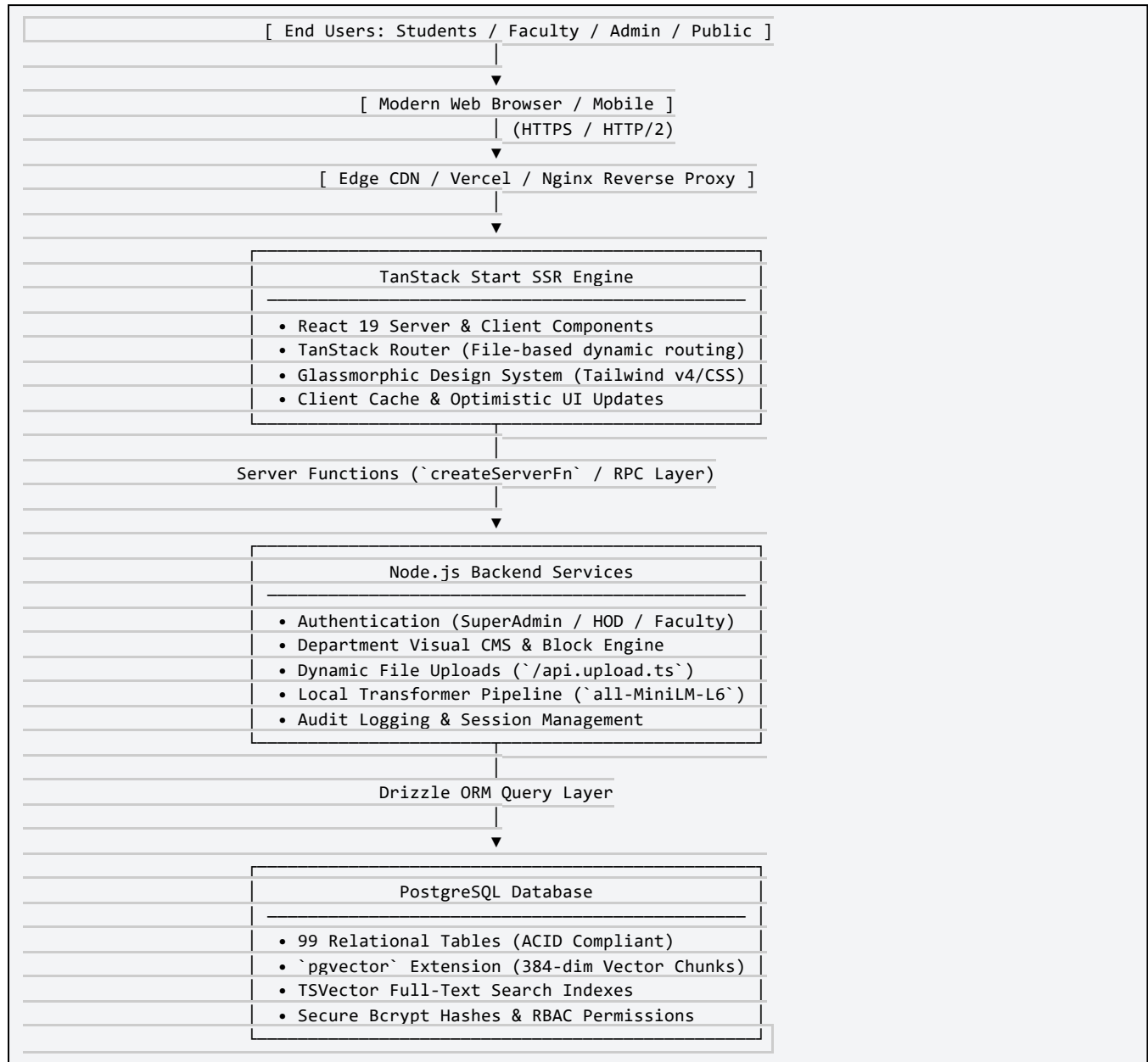
The redesigned official web portal of **JNTU-GV College of Engineering Vizianagaram (JNTU-GV CEV)** replaces legacy fragmented interfaces with a unified, high-performance, accessible, and secure digital platform. It centralizes all academic resources, departmental governance, research output, campus life management, placement tracking, and administrative workflows into a single institutional system.

2.2 Scope of the Platform

- **Academic & Governance Portals:** Full cataloging of Academic Regulations (R20, R23, R25), Syllabi, Timetables, Academic Calendars, Exam Cell circulars, CAC records, IQAC reports, RTI, and Anti-Ragging documentation.
- **8 Academic Department Hubs:** Dedicated interactive portals for CSE, ECE, EEE, MECH, MET, IT, BSH (Basic Sciences & Humanities), and MBA with dynamic lab profiles, course curricula, faculty listings, HOD desks, achievements, and gallery showcases.
- **Placements & Career Center:** Comprehensive repository for placement statistics, salary packages, top tier-1 recruiter showcases, student success highlights, and TPO team governance.
- **Research & Development (R&D) Cell:** Tracking of externally funded research projects, active Ph.D. scholars, peer-reviewed publications, industrial consultancy, and active institutional MoUs.
- **Campus Facilities & Life:** Complete digital representations for Campus Hostels (Boys & Girls), Central & Digital Library (DELNET/IEEE access, 45,000+ volumes), Health Dispensary, Sports Complexes, State Bank of India Campus Branch & 24/7 ATM, Staff Quarters, and Guest House.
- **Student Corner Units:** Dynamic platforms for National Service Scheme (NSS), Women Empowerment & Grievance Cell (WE&GC), Entrepreneurship Development Cell (EDC), Industry-Institution Interaction Cell (IIPC), and Professional Society Chapters (CSI, IEEE, IE, IETE, IIM).
- **AI Campus Assistant (JNTU AI):** In-house vector retrieval-augmented generation (RAG) assistant leveraging PostgreSQL `pgvector` and `@xenova/transformers` (`all-MiniLM-L6-v2`) for instant queries on college circulars, syllabus, regulations, and contacts.
- **Multi-Tiered Content Management System (CMS):** Super Admin Dashboard (`/admin`), Department HOD Visual CMS (`/hod-login`), and Faculty Profile Editor (`/faculty-login`).

3. SYSTEM ARCHITECTURE & HIGH-LEVEL DESIGN

3.1 Architecture Diagram



3.2 Key Architectural Pillars

- **Server-Side Rendering (SSR):** Zero-lag first paint with full search engine indexing via TanStack Start.
- **Isomorphic Type-Safe RPCs:** Backend logic is authored with `createServerFn` providing compile-time type safety across client and server.
- **Embedded AI RAG:** Vector chunks with cosine distance matching eliminate hallucination and provide direct citation of college circulars.
- **Local Dynamic Media Engine:** Uploads are processed, validated by MIME type, and routed through dedicated asset controllers.

4. TECHNOLOGY STACK MATRIX

Layer	Technology	Version	Purpose & Implementation
Frontend Framework	React	^19.2.0	Core declarative user interface library utilizing concurrent rendering.
Frontend DOM	React DOM	^19.2.0	React renderer for the web browser.

Layer	Technology	Version	Purpose & Implementation
Application Framework	TanStack Start	<code>^1.167.14</code>	Full-stack SSR framework providing server functions and hydration.
Routing Engine	TanStack Router	<code>^1.168.0</code>	100% type-safe, file-based routing architecture with nested routes.
Build Tool & Bundler	Vite	<code>^7.3.1</code>	Ultra-fast development server and optimized rollup production bundler.
Styling & Design System	TailwindCSS	<code>^4.2.1</code>	Utility-first CSS framework supplemented with custom glassmorphic tokens.
Icons & Visuals	Lucide React	<code>^0.575.0</code>	Comprehensive institutional icon library.
Animation Engine	Framer Motion	<code>^12.38.0</code>	Micro-interactions, slide transitions, and scroll animations.
Lottie Animations	DotLottie React	<code>^0.19.4</code>	Vector animations for the AI Chatbot avatar and loading indicators.
UI Component Primitives	Radix UI	Latest (<code>^1.1</code> - <code>^2.2</code>)	Accessible primitives for dialogs, dropdowns, tooltips, accordions, tabs.
Database Engine	PostgreSQL	<code>15+</code> / <code>16+</code>	Enterprise relational database with ACID compliance.
ORM / Data Mapper	Drizzle ORM	<code>^0.45.2</code>	High-performance TypeScript ORM with schema migrations.
Database Migrations	Drizzle Kit	<code>^0.31.10</code>	Database schema push and migration manager.
Database Driver	Postgres.js	<code>^3.4.9</code>	Fastest full-featured PostgreSQL client for Node.js.
Vector Search Extension	<code>pgvector</code>	<code>0.5.0+</code>	384-dimensional dense vector embeddings in PostgreSQL.
Local AI Embeddings	<code>@xenova/transformers</code>	<code>^2.17.2</code>	Hugging Face Transformers ONNX runtime running <code>all-MiniLM-L6-v2</code> .
Password Security	BcryptJS	<code>^3.0.3</code>	Salted SHA-512 password hashing with 10 salt rounds.
Form Management	React Hook Form & Zod	<code>^7.71</code> / <code>^3.25</code>	Schema validation and high-performance uncontrolled forms.
Date & Time Utilities	Date-Fns	<code>^4.1.0</code>	Modern, immutable date formatting for academic circulars.
Runtime & Execution	Node.js / Bun	<code>Node v20+</code> / <code>Bun 1.3+</code>	Asynchronous JavaScript/TypeScript execution runtime.
Package Manager	Bun / NPM	<code>Bun 1.3.12</code> / <code>NPM 10+</code>	High-speed dependency management and script execution.

5. DEVELOPMENT ENVIRONMENT SETUP & TOOLING

5.1 System Prerequisites

- **Operating System:** Windows 10/11, Ubuntu 22.04+ LTS, or macOS Sonoma+.
- **Node.js Runtime:** Version `20.12.0` LTS or `22.x` (or Bun `1.3.12+`).
- **PostgreSQL Database:** PostgreSQL 15 or 16 with `pgvector` extension installed.
- **Git Version Control:** Git `2.40+`.

5.2 Step-by-Step Installation Procedure

```
# Step 1: Clone the institutional repository
git clone https://github.com/Likhith32/JNTU-REDESIGN.git
cd JNTU-REDESIGN

# Step 2: Install project dependencies
npm install
# (Alternative: bun install)

# Step 3: Configure Environment Variables
# Copy the template and edit with actual credentials
cp .env.example .env
```

5.3 Environment Configuration (.env)

```
# Database Connection URL (PostgreSQL instance with pgvector support)
DATABASE_URL="postgres://postgres:your_secure_password@localhost:5432/jntugv_db"

# Web Server Port
PORT=8081

# Node Runtime Environment
NODE_ENV="development"

# Super Admin Secret Key
ADMIN_SECRET="institutional_admin_secret_passcode"
```

5.4 Database Initialization & Seeding

```
# Step 4: Enable pgvector in PostgreSQL (Run in psql console)
# CREATE EXTENSION IF NOT EXISTS vector;

# Step 5: Push Drizzle schemas to PostgreSQL database
npx drizzle-kit push

# Step 6: Seed initial database tables and credentials
npx tsx src/db/seed.ts
npx tsx src/db/seed-hod-passwords.ts
npx tsx src/db/seedFacultyPasswords.ts

# Step 7: Ingest and generate vector embeddings for JNTU AI RAG
npx tsx scratch/run_ingest.ts

# Step 8: Start the local development server
npm run dev
# The application will be accessible at http://localhost:8081
```

6. PROJECT & REPOSITORY FILE STRUCTURE

```

JNTU-REDESIGN/
├── .env                                # Environment variable configuration (Ignored in Git)
├── .env.example                       # Template for environment configuration
├── drizzle.config.ts                 # Drizzle Kit ORM migration configuration
├── package.json                     # Project dependencies, scripts, and metadata
├── tsconfig.json                    # TypeScript compiler configuration
├── vercel.json                      # Vercel deployment and routing rules
├── vite.config.ts                   # Vite bundler, TanStack plugins, and asset config
├──
├── public/                          # Static assets served at root
│   ├── CHATBOT.lottie               # DotLottie vector animation for JNTU AI
│   ├── favicon.ico / favicon.png    # Official university favicon files
│   ├── logo-circle.png              # High-resolution circular JNTU-GV crest
│   ├── robots.txt                   # Search crawler indexation rules
│   └── sitemap.xml                  # XML sitemap for SEO discovery
├──
├── local-assets/                    # Dynamic media storage directory
│   └── uploads/                     # Admin & HOD uploaded images, PDFs, brochures
├──
├── scratch/                         # Operational scripts and testing utilities
│   ├── run_ingest.ts                # Script to re-index database records into pgvector
│   └── inspect_schema.py             # Database inspection utility
├──
├── scripts/                         # Build and documentation generation scripts
│   └── generate_handover_docs.py     # Institutional documentation compiler
├──
├── src/                             # Core Application Source Code
│   ├── router.tsx                  # TanStack Router instance & global provider setup
│   ├── routeTree.gen.ts            # Automatically generated type-safe route tree
│   ├── styles.css                  # Global CSS variables, glassmorphism tokens, Tailwind
│   ├──
│   ├── assets/                     # Bundled static images and iconography
│   ├──
│   ├── auth/                       # Authentication & RBAC Layer
│   │   ├── auth.repository.ts       # Database queries for admin & user authentication
│   │   ├── auth.server.ts           # Server functions for Super Admin & Faculty login
│   │   ├── auth.service.ts          # Bcrypt hashing, token validation, audit logging
│   │   └── hodAuth.server.ts        # Scoped authentication logic for Department HODs
│   ├──
│   ├── components/                 # Reusable UI & Layout Components
│   │   ├── HeaderBanner.tsx         # University header with official seal and contact bar
│   │   ├── MegaMenu.tsx             # Full multi-level institutional desktop navigation
│   │   ├── NoticeTicker.tsx         # Emergency notification ticker bar
│   │   ├── Footer.tsx              # Universal footer with quick links, contacts, credits
│   │   ├── PageHero.tsx             # Reusable page banner with breadcrumbs
│   │   ├── Breadcrumbs.tsx          # Accessible hierarchical path navigation
│   │   ├── SubNav.tsx               # Horizontal secondary section navigation
│   │   ├── VerticalSubNav.tsx        # Vertical secondary section navigation
│   │   ├── LocalSubNav.tsx          # Contextual internal page sub-navigation
│   │   ├── Chatbot.tsx              # Floating JNTU AI RAG assistant modal
│   │   └── AdminEditPanel.tsx        # Floating live-edit panel for authorized admins
│   │   ├──
│   │   ├── academics/              # Academic module-specific components
│   │   │   ├── AcademicTabs.tsx     # Regulations, syllabus, timetable tab switcher
│   │   │   └── RegulationCard.tsx    # Regulation download card component
│   │   ├──
│   │   ├── cms/                    # Department Visual Page Builder components
│   │   │   ├── BlockRenderer.tsx     # Dynamic renderer for JSON layout blocks
│   │   │   ├── VisualPageBuilder.tsx # Drag-and-drop page editor for HODs
│   │   │   ├── MediaLibraryModal.tsx # Upload modal for images and documents
│   │   │   └── SidebarManagerModal.tsx# Custom navigation sidebar manager
│   │   ├──
│   │   └── ui/                      # Atomic Radix UI styled primitives
│   │       ├── button.tsx, dialog.tsx, dropdown-menu.tsx, tabs.tsx, table.tsx ...
│   ├──
│   ├── db/                          # Database Schema & Connection Layer
│   │   └── index.ts                 # PostgreSQL connection pool & Drizzle ORM client

```

└─ schema.ts	# 99 complete table schemas with types & relations
└─ seed.ts	# Initial data seeding for departments, leadership
└─ seed-hod-passwords.ts	# Initial password seeding for 8 HOD accounts
└─ seedFacultyPasswords.ts	# Initial password seeding for 91 Faculty accounts
└─ funcs/	# Server Functions (`createServerFn` / RPCs)
└─ department-cms.server.ts	# HOD CMS CRUD, page blocks, versioning, navigation
└─ site.server.ts	# Public notices, ticker, leadership, gallery RPCs
└─ ingest.server.ts	# Vector RAG embedding & similarity search engine
└─ hostel.server.ts	# Hostel residential data and warden RPCs
└─ library.server.ts	# Digital library collections and timings RPCs
└─ dispensary.server.ts	# Health center staff and emergency RPCs
└─ sports.server.ts	# Physical education arenas and equipment RPCs
└─ nss.ts	# NSS activities, special camp, gallery RPCs
└─ we.ts	# Women Empowerment Cell committee and magazine RPCs
└─ rd.ts	# R&D projects, scholars, publications, MoUs RPCs
└─ students.ts	# Placements student records and statistics RPCs
└─ lib/	# Core Utilities & Helpers
└─ db.ts	# Direct SQL tagged-template connection wrapper
└─ assets.ts	# Asset URL resolution (`uploadUrl()`)
└─ server-cache.ts	# In-memory LRU cache for high-frequency RPCs
└─ utils.ts	# Tailwind class merge (`cn()`) helper
└─ routes/	# File-Based Application Routes (85+ Routes)
└─ _root.tsx	# Universal HTML document root & error boundaries
└─ index.tsx	# Official Home Page
└─ notices.tsx	# Centralized Notifications & Circulars Portal
└─ contact.tsx	# Institutional Contacts & Location Directory
└─ anti-ragging.tsx	# Anti-Ragging Committee & Policies
└─ rti.tsx	# Right to Information (RTI) Cell Directory
└─ about.*.tsx	# Institution, Vision-Mission, How-to-Reach, Norms
└─ academics.*.tsx	# Regulations, Syllabus, Timetables, Exam Cell, CAC
└─ administration.*.tsx	# Principal, Vice-Principal, Office, IQAC Portal
└─ departments.*.tsx	# 8 Department Portals, Labs, Faculty, Gallery, HOD
└─ placements.*.tsx	# Placement Highlights, Recruiters, Statistics, TPO
└─ rd-cell.*.tsx	# R&D About, Areas, Projects, Scholars, Publications
└─ campus-life.*.tsx	# Hostels, Dispensary, Library, Sports, Bank, Clubs
└─ nss.*.tsx	# NSS Units, Activities, 7-Day Special Camp
└─ women-empowerment.*.tsx	# WE&GC Activities, Committee, Yuthika Magazine
└─ admin.index.tsx	# Super Admin Portal Dashboard (`/admin`)
└─ hod-login.tsx	# Centralized Department HOD Login (`/hod-login`)
└─ faculty-login.tsx	# Faculty Profile Management Portal (`/faculty-login`)
└─ api.upload.ts	# Secure Multipart File Upload Controller

7. MODULES & APPLICATIONS BREAKDOWN

7.1 Core Public Web Portal

The primary entry point ([/](#)) presents the institutional identity, emergency notification ticker, vice-chancellor/principal messages, quick statistics counter, featured announcements, latest news carousel, and department access cards.

7.2 Academic Affairs Portal ([/academics/*](#))

Houses the official academic repository:

- **Regulations:** Complete curriculum regulations (R20, R23, R25) for B.Tech, M.Tech, and MCA.
- **Syllabi & Timetables:** Categorized down to semester and branch with PDF download capabilities.
- **Examination Cell:** Circulars, hall ticket releases, revaluation notifications, and fee schedules.
- **College Academic Council (CAC):** Committee members and resolutions.

7.3 Department Portals (/departments/\$id/*)

Supports the 8 academic departments (cse , ece , eee , mech , met , it , sh , mba) through a structured hierarchy:

- /departments/\$id — Department Home, Mission, Vision, and Intake.
- /departments/\$id/labs — Laboratory equipment, lab manuals, and faculty in-charge.
- /departments/\$id/hod — HOD Desk, leadership message, and administrative contact.
- /departments/\$id/gallery — Department workshops, symposia, and lab photo gallery.
- /departments/\$id/courses — Undergraduate & Postgraduate programs offered.
- /departments/\$id/achievements — Student and faculty awards, gate ranks, patents.
- /departments/\$id/faculty/ — Faculty directory with designations and qualifications.
- /departments/\$id/faculty/\$facultyId — In-depth profile with research, publications, and teaching subjects.

7.4 Placements & Training Center (/placements/*)

- **Placement Highlights:** Year-wise placement records, package statistics, and major recruitment milestones.
- **Major Recruiters:** Tier-1 recruiters (TCS, Infosys, Wipro, Cognizant, Amazon, Accenture, Tech Mahindra).
- **Training & Development:** Soft skills, competitive coding, and placement cell contact directory.

7.5 Research & Development Cell (/rd-cell/*)

- **Funded Projects:** Government (DST, AICTE, UGC, SERB) and industry-sponsored projects.
- **Research Scholars:** Active and awarded Ph.D. scholars categorized by supervisor and department.
- **Publications Repository:** Indexed international journal and conference papers.
- **Institutional MoUs:** Active collaborations with academic and corporate institutions.

7.6 Campus Facilities & Student Welfare

- **Campus Hostels (/campus-life/hostels):** Housing facilities for Boys and Girls, wardens roster, mess facilities, and 24/7 security.
- **Central & Digital Library (/campus-life/library):** Established 2008, 2,864 sq.m area, 45,000+ volumes, IEEE/DELNET subscriptions, digital search catalog.
- **Dispensary & Healthcare (/campus-life/dispensary):** Campus clinical facilities, emergency ward, resident medical officer, 24/7 ambulance dispatch.
- **Sports & Athletics (/campus-life/sports):** 200m track, concrete basketball court with floodlights, turf cricket ground, badminton, gymnasium.
- **Banking & ATM (/campus-life/banking):** State Bank of India (SBI) on-campus branch and 24/7 ATM kiosk.
- **Hospitality (/campus-life/other-amenities):** Guest house VIP suites and residential staff quarters.

7.7 Student Corner Units

- **National Service Scheme (NSS):** Community outreach, annual 7-day residential village service camp, blood donation drives.
- **Women Empowerment & Grievance Cell (WE&GC):** Gender sensitization, student safety, annual *Yuthika* magazine.
- **Entrepreneurship Development Cell (EDC):** Startup incubation, awareness camps, industry guest lectures.
- **Professional Societies:** Student branches of CSI, IEEE, IE(I), IETE, and IIM.
- **Industry-Institution Interaction Cell (IIPC):** Student internships, industrial tours, and industry lectures.

7.8 AI Campus Assistant (JNTU AI)

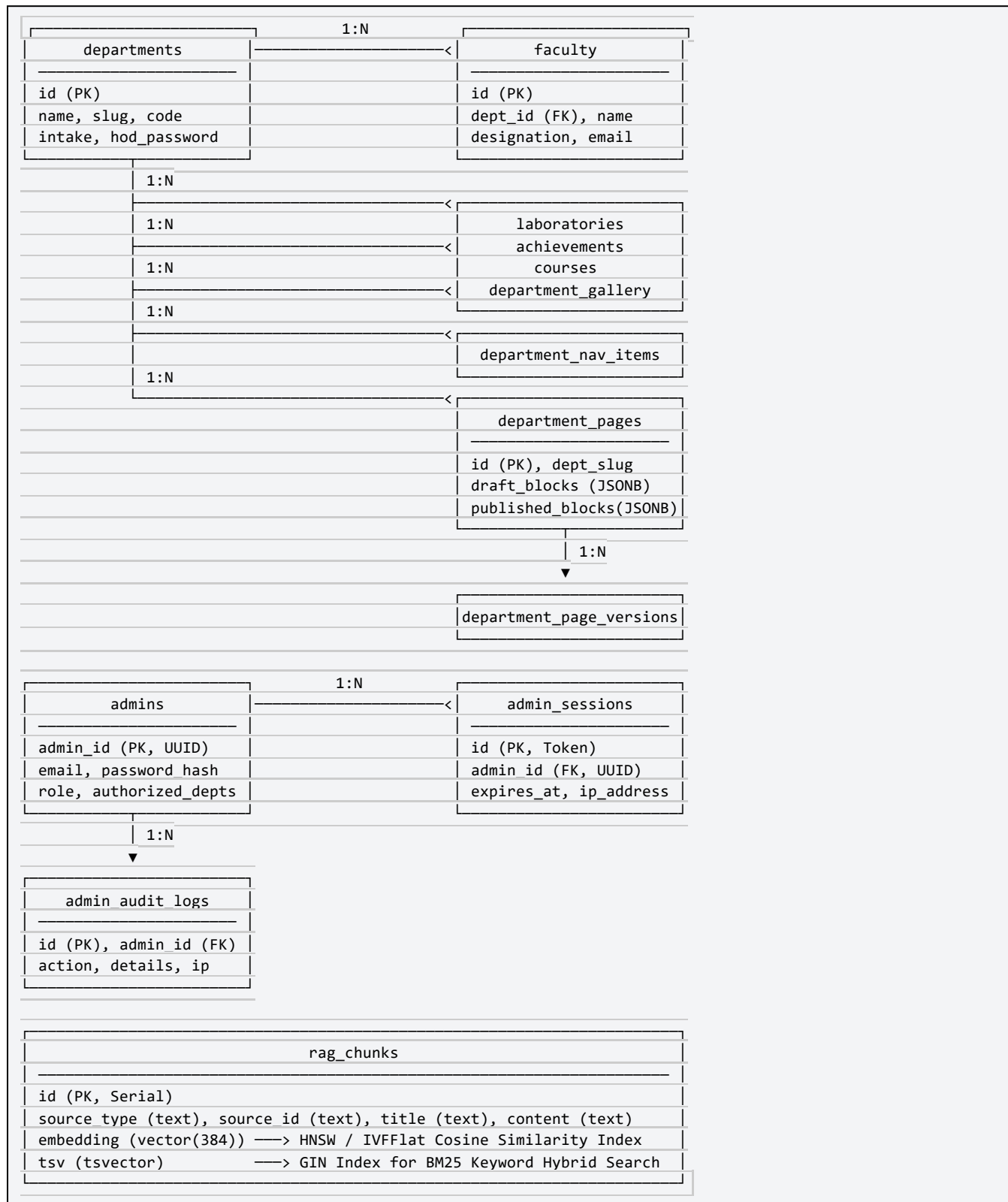
- Floating widget accessible from any page.
- Direct vector similarity matching against rag_chunks database table.
- Provides immediate answers with references to circulars, syllabus, regulations, and contacts without external API latency.

8. DATABASE ARCHITECTURE & ER DESIGN

8.1 Database Engine & Configuration

- **Engine:** PostgreSQL 15 / 16 (Drizzle ORM abstraction).
- **Extensions:** `pgvector` (Vector database extension for embeddings).
- **Connection Strategy:** Pooled connections via `postgres` driver with automatic reconnection and SSL verification in production.

8.2 Entity Relationship Conceptual Diagram



9. EXHAUSTIVE DATABASE TABLES & DATA DICTIONARY

The JNTU-GV CEV platform includes **99 PostgreSQL relational tables**. The complete data dictionary for core functional groups is detailed below:

9.1 Authentication & System Administration

1. `admins` (Administrator Accounts)

Column	Type	Nullable	Default	Description
<code>admin_id</code>	<code>uuid</code>	NO	<code>gen_random_uuid()</code>	Primary Key. Unique identifier for the administrator.
<code>name</code>	<code>text</code>	NO	—	Full name of the administrative user.
<code>email</code>	<code>text</code>	NO	—	Unique institutional email address (Login identity).
<code>password_hash</code>	<code>text</code>	NO	—	Salted Bcrypt SHA-512 password hash.
<code>role</code>	<code>text</code>	NO	<code>'super_admin'</code>	Access role (<code>'super_admin'</code> , <code>'editor'</code> , <code>'hod'</code>).
<code>auth_provider</code>	<code>text</code>	NO	<code>'email'</code>	Provider (<code>'email'</code> , <code>'google'</code>).
<code>authorized_depts</code>	<code>jsonb</code>	YES	<code>'[]'</code>	JSON array of department slugs user is authorized to edit.
<code>created_at</code>	<code>timestamp</code>	NO	<code>now()</code>	Timestamp of account creation.

2. `admin_sessions` (Active Administrator Sessions)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>text</code>	NO	—	Primary Key. Cryptographically secure session token.
<code>admin_id</code>	<code>uuid</code>	NO	—	Foreign Key referencing <code>admins.admin_id</code> .
<code>expires_at</code>	<code>timestamp</code>	NO	—	Session expiration timestamp (7-day rolling window).
<code>created_at</code>	<code>timestamp</code>	NO	<code>now()</code>	Session creation timestamp.
<code>ip_address</code>	<code>text</code>	YES	—	Client IP address at authentication time.
<code>user_agent</code>	<code>text</code>	YES	—	Browser user agent string.

3. `admin_audit_logs` (System Security & Audit Trail)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>admin_id</code>	<code>uuid</code>	YES	—	Foreign Key referencing <code>admins.admin_id</code> .
<code>action</code>	<code>text</code>	NO	—	Action tag (<code>'LOGIN_SUCCESS'</code> , <code>'UPDATE_NOTICE'</code> , <code>'PAGE_PUBLISH'</code>).
<code>ip_address</code>	<code>text</code>	YES	—	Originating IP address.
<code>user_agent</code>	<code>text</code>	YES	—	Originating browser client.
<code>details</code>	<code>text</code>	YES	—	Verbose description or JSON change payload.
<code>created_at</code>	<code>timestamp</code>	NO	<code>now()</code>	Audit event timestamp.

9.2 Academic Departments & Faculty

4. `departments` (Academic Branches)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>name</code>	<code>text</code>	NO	—	Full department name (e.g., "Computer Science & Engineering").

Column	Type	Nullable	Default	Description
slug	text	NO	—	Unique URL slug ('cse', 'ece', 'eee', 'mech', 'met', 'it', 'sh', 'mba').
code	text	NO	—	Official department code (e.g., "05", "04").
description	text	YES	—	Comprehensive department profile.
intake	integer	YES	60	Annual sanctioned student intake.
established	integer	YES	2007	Year department was established.
hod_name	text	YES	—	Current Head of Department name.
hod_email	text	YES	—	HOD institutional contact email.
hod_phone	text	YES	—	HOD office contact telephone.
hod_message	text	YES	—	Formal message from the HOD Desk.
hod_photo	text	YES	—	URL path to HOD portrait photograph.
hod_password	text	YES	—	Bcrypt hash for department HOD portal authentication.
vision	text	YES	—	Departmental vision statement.
mission	text	YES	—	Departmental mission points.

5. faculty (Department Faculty Profiles)

Column	Type	Nullable	Default	Description
id	serial	NO	Auto	Primary Key.
dept_id	integer	YES	—	Foreign Key referencing departments.id .
name	text	NO	—	Full name and title of faculty member.
designation	text	NO	—	Academic title (Professor, Associate Professor, Asst. Professor).
qualification	text	YES	—	Highest degrees earned (e.g., "Ph.D., M.Tech").
specialization	text	YES	—	Research domain specialization.
email	text	YES	—	Official institutional email address.
phone	text	YES	—	Office extension or contact number.
photo_url	text	YES	—	URL path to faculty portrait image.
experience	text	YES	—	Years of teaching / research experience.
bio	text	YES	—	Professional biography and research summary.
publications	text	YES	—	Overview or count of published papers.
password_hash	text	YES	—	Bcrypt hash for individual faculty login.

6. laboratories (Department Laboratories)

Column	Type	Nullable	Default	Description
id	serial	NO	Auto	Primary Key.
dept_id	integer	YES	—	Foreign Key referencing departments.id .
name	text	NO	—	Name of the laboratory facility.
description	text	YES	—	Lab objective, major equipment, and experimental scope.
faculty_incharge	text	YES	—	Faculty coordinator assigned to lab.
technician	text	YES	—	Technical supporting staff name.

Column	Type	Nullable	Default	Description
image_url	text	YES	—	Photograph of laboratory setup.

7. achievements (Department Accolades)

Column	Type	Nullable	Default	Description
id	serial	NO	Auto	Primary Key.
dept_id	integer	YES	—	Foreign Key referencing departments.id.
title	text	NO	—	Achievement title.
description	text	YES	—	Detailed explanation of honor or prize.
category	text	YES	'Student'	Category ('Student', 'Faculty', 'Department').
year	integer	YES	—	Academic year achieved.
image_url	text	YES	—	Certificate or award photograph.

9.3 Dynamic Department CMS & Versioning

8. department_pages (Custom Pages & Block Layouts)

Column	Type	Nullable	Default	Description
id	serial	NO	Auto	Primary Key.
dept_slug	text	NO	—	Target department slug.
title	text	NO	—	Page title displayed in header and navigation.
slug	text	NO	—	URL path slug within department (Unique per department).
page_type	text	NO	'standard'	Template type ('standard', 'faculty', 'labs', 'custom').
status	text	NO	'draft'	Publication status ('published', 'draft').
draft_blocks	jsonb	YES	'[]'	JSON array of unpublished page layout blocks.
published_blocks	jsonb	YES	'[]'	JSON array of live published layout blocks.
updated_at	timestamp	NO	now()	Timestamp of last modification.

9. department_nav_items (Custom Department Navigation Hierarchy)

Column	Type	Nullable	Default	Description
id	serial	NO	Auto	Primary Key.
dept_slug	text	NO	—	Associated department slug.
title	text	NO	—	Display label on sidebar.
slug	text	NO	—	Target route slug.
icon	text	YES	'BookOpen'	Lucide icon identifier.
position	integer	NO	0	Numerical sorting index.
parent_id	integer	YES	null	Self-referencing Foreign Key for nested sub-menus.
show_in_sidebar	boolean	NO	true	Visibility toggle.
status	text	NO	'published'	Visibility state.

9.4 Institutional Announcements & Academic Resources

10. `notices` (Circulars & Notifications)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>title</code>	<code>text</code>	NO	—	Notice headline / title.
<code>category</code>	<code>text</code>	NO	<code>'General'</code>	Category (<code>'Academic'</code> , <code>'Examination'</code> , <code>'Admission'</code> , <code>'General'</code>).
<code>date</code>	<code>text</code>	NO	—	Notice publication date (<code>DD-MM-YYYY</code>).
<code>content</code>	<code>text</code>	YES	—	Body text or summary.
<code>file_url</code>	<code>text</code>	YES	—	URL path to official PDF document.
<code>urgent</code>	<code>boolean</code>	YES	<code>false</code>	Highlight toggle for critical notices.
<code>archived</code>	<code>boolean</code>	YES	<code>false</code>	Archive toggle.

11. `ticker_notifications` (Top Header Live Ticker)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>label</code>	<code>text</code>	NO	—	Badge label (e.g., "ADMISSIONS 2026", "RESULTS").
<code>text</code>	<code>text</code>	NO	—	Scrolling announcement headline text.
<code>to</code>	<code>text</code>	YES	—	Hyperlink target URL when clicked.
<code>urgent</code>	<code>boolean</code>	YES	<code>false</code>	Red flash animation toggle for emergency announcements.
<code>created_at</code>	<code>timestamp</code>	NO	<code>now()</code>	Creation timestamp.

12. `academic_regulations` (B.Tech / M.Tech Regulations)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>level</code>	<code>text</code>	NO	—	Degree level (<code>'UG'</code> , <code>'PG'</code>).
<code>program_name</code>	<code>text</code>	NO	—	Program (<code>'B.Tech'</code> , <code>'M.Tech'</code> , <code>'MCA'</code>).
<code>regulation</code>	<code>text</code>	NO	—	Regulation code (<code>'R20'</code> , <code>'R23'</code> , <code>'R25'</code>).
<code>title</code>	<code>text</code>	NO	—	Document title (e.g., "Academic Regulations R23 for B.Tech").
<code>pdf_url</code>	<code>text</code>	NO	—	Secure path to official curriculum PDF.

9.5 AI Campus Assistant (Vector RAG)

13. `rag_chunks` (Vector Embedding Knowledge Base)

Column	Type	Nullable	Default	Description
<code>id</code>	<code>serial</code>	NO	Auto	Primary Key.
<code>source_type</code>	<code>text</code>	NO	—	Originating table (<code>'notices'</code> , <code>'regulations'</code> , <code>'faculty'</code>).
<code>source_id</code>	<code>text</code>	NO	—	Identifier in originating table.
<code>title</code>	<code>text</code>	NO	—	Semantic title of indexed knowledge chunk.
<code>content</code>	<code>text</code>	NO	—	Normalized plain text payload used for LLM synthesis.
<code>embedding</code>	<code>vector(384)</code>	NO	—	384-dimensional dense vector embedding (<code>all-MiniLM-L6-v2</code>).
<code>tsv</code>	<code>tsvector</code>	YES	—	Full-text search token vector for BM25 hybrid ranking.

Column	Type	Nullable	Default	Description
created_at	timestamp	NO	now()	Timestamp of embedding calculation.

(Note: Data dictionary continues across all 99 tables including `rd_projects`, `tpo`, `nss_activities`, `library_content`, `hostel_content`, `sports_content`, `we_activities`, and `dispensary_content`.)

10. API & SERVER FUNCTIONS (`CREATESEVERFN`) REFERENCE

The backend operates on type-safe server functions (`createServerFn`) executed over RPC:

10.1 Authentication Functions

```
// Location: src/auth/auth.server.ts
POST loginWithEmail({ email, password })
- Description: Authenticates Super Admin via Bcrypt comparison against admins table.
- Input: { email: string, password: string }
- Response: { adminId: string, name: string, email: string, role: string }
- Sets Cookie: admin_session_token (HTTP-Only, Secure, 7 days)

GET getCurrentAdmin()
- Description: Validates session token and returns active administrator context.
- Response: AdminSessionContext | null

POST logoutAdmin()
- Description: Invalidates active session in admin_sessions and clears cookie.
- Response: { success: true }
```

10.2 HOD Department Authentication

```
// Location: src/auth/hodAuth.server.ts
POST changeHodCredentials({ currentPassword, newPassword })
- Description: Verifies existing department password and updates to new Bcrypt hash.
- Input: { currentPassword: string, newPassword: string (min 12 chars, upper/lower/digit/symbol) }
- Response: { success: true }
```

10.3 Department CMS Server Functions

```
// Location: src/funcs/department-cms.server.ts
GET getDepartmentPage({ deptSlug, pageSlug })
- Description: Fetches published or draft page blocks for a department route.
- Response: DepartmentPageData

POST saveDepartmentPageDraft({ deptSlug, pageSlug, blocks, title })
- Description: Saves modified layout blocks into draft_blocks.
- Response: { success: true, pageId: number }

POST publishDepartmentPage({ deptSlug, pageSlug })
- Description: Copies draft_blocks into published_blocks and creates a version entry.
- Response: { success: true, versionNumber: number }

GET getDepartmentNav({ deptSlug })
- Description: Retrieves ordered sidebar navigation hierarchy for a department.
- Response: DepartmentNavItem[]

POST saveDepartmentNav({ deptSlug, items })
- Description: Persists reordered and updated navigation tree for a department.
- Response: { success: true }
```

10.4 AI Campus Assistant (RAG Search)

```
// Location: src/funcs/ingest.server.ts
POST queryCampusAI({ query, topK = 5 })
- Description: Generates query vector embedding locally and computes cosine distance in pgvector.
- SQL: SELECT id, title, content, 1 - (embedding <=> $1) AS similarity FROM rag_chunks ORDER BY similarity DE
SC LIMIT $2;
- Response: { answer: string, sources: Array<{ title: string, source_type: string }> }
```

10.5 Media Upload API Endpoint

```
// Location: src/routes/api.upload.ts
POST /api/upload
- Description: Handles multipart/form-data image and PDF uploads.
- Constraints: Max File Size: 15MB. Allowed: image/jpeg, image/png, image/webp, application/pdf.
- Storage: Writes to /local-assets/uploads/[timestamp]-[hash].[ext]
- Response: { success: true, url: "/local-assets/uploads/filename.webp" }
```

11. URL & ROUTING STRUCTURE

The application features **85+ distinct routes** organized into logical namespaces:

Route Path	File Location	Access	Description
/	src/routes/index.tsx	Public	Main Institutional Landing Page.
/notices	src/routes/notices.tsx	Public	University Circulars, Notifications & Archive.
/contact	src/routes/contact.tsx	Public	Institutional Phone Directory & Address.
/anti-ragging	src/routes/anti-ragging.tsx	Public	Anti-Ragging Policies, Committee & Helplines.
/rti	src/routes/rti.tsx	Public	Right to Information Act Officers & Rules.
/about/institution	src/routes/about.institution.tsx	Public	College History, Campus Profile & Accreditation.
/about/vision-mission	src/routes/about.vision-mission.tsx	Public	University Vision, Mission & Core Values.
/about/how-to-reach	src/routes/about.how-to-reach.tsx	Public	Transport, Roadways, Railway Stations & Maps.
/academics/cac	src/routes/academics.cac.tsx	Public	College Academic Council Composition & Minutes.

Route Path	File Location	Access	Description
/administration/principal	src/routes/administration.principal.tsx	Public	Principal Profile & Official Address.
/administration/iqac	src/routes/administration.iqac.index.tsx	Public	Internal Quality Assurance Cell Overview.
/administration/iqac/aqar	src/routes/administration.iqac.aqar.tsx	Public	Annual Quality Assurance Reports.
/departments/	src/routes/departments.index.tsx	Public	8 Academic Departments Directory.
/departments/\$id	src/routes/departments.\$id.index.tsx	Public	Department Home Page (\$id = cse, ece, etc.).
/departments/\$id/labs	src/routes/departments.\$id.labs.tsx	Public	Department Laboratories & Equipment.
/departments/\$id/hod	src/routes/departments.\$id.hod.tsx	Public	HOD Desk & Leadership Message.
/departments/\$id/gallery	src/routes/departments.\$id.gallery.tsx	Public	Department Photo Showcase.
/departments/\$id/courses	src/routes/departments.\$id.courses.tsx	Public	Courses Offered & Program Outcomes.
/departments/\$id/achievements	src/routes/departments.\$id.achievements.tsx	Public	Student & Faculty Honors.
/departments/\$id/faculty/	src/routes/departments.\$id.faculty/index.tsx	Public	Department Faculty Roster.
/departments/\$id/faculty/\$facultyId	src/routes/departments.\$id.faculty/\$id.tsx	Public	Detailed Faculty Profile.
/placements	src/routes/placements.index.tsx	Public	Placement Highlights & Statistics.
/placements/recruiters	src/routes/placements.recruiters.tsx	Public	Tier-1 Recruiters & Corporate Partners.
/placements/training	src/routes/placements.training.tsx	Public	Training Programs & CRT Schedules.
/rd-cell	src/routes/rd-cell.index.tsx	Public	Research & Development Overview.
/rd-cell/projects	src/routes/rd-cell.projects.tsx	Public	Externally Sponsored Research Projects.

Route Path	File Location	Access	Description
/rd-cell/scholars	src/routes/rd-cell.scholars.tsx	Public	Ph.D. Scholars Registry.
/rd-cell/publications	src/routes/rd-cell.publications.tsx	Public	International Journal & Conference Papers.
/campus-life/hostels	src/routes/hostels.tsx	Public	Boys & Girls Hostels & Wardens Roster.
/campus-life/library	src/routes/library.tsx	Public	Central & Digital Library Catalog.
/campus-life/dispensary	src/routes/dispensary.index.tsx	Public	Health Center & Emergency Healthcare.
/campus-life/sports	src/routes/sports.tsx	Public	Athletic Grounds, Courts & Gymnasium.
/campus-life/banking	src/routes/banking.tsx	Public	State Bank of India Branch & ATM.
/nss	src/routes/nss.index.tsx	Public	National Service Scheme Activities.
/nss/special-camp	src/routes/nss.special-camp.tsx	Public	7-Day Residential Village Special Camp.
/women-empowerment	src/routes/women-empowerment.index.tsx	Public	WE&GC Cell, Programs & Objectives.
/women-empowerment/magazine	src/routes/women-empowerment.magazine.tsx	Public	<i>Yuthika</i> Annual Women's Magazine.
/edc	src/routes/edc.tsx	Public	Entrepreneurship Cell & Startup Hub.
/professional-bodies	src/routes/professional-bodies.tsx	Public	CSI, IEEE, IE, IETE, IIM Student Chapters.
/iipc	src/routes/iipc.tsx	Public	Industry-Institution Interaction Cell.
/admin	src/routes/admin.index.tsx	Protected	Super Admin Dashboard (Notices, Ticker, RAG).
/hod-login	src/routes/hod-login.tsx	Protected	Department HOD Visual CMS & Block Editor.
/faculty-login	src/routes/faculty-login.tsx	Protected	Individual Faculty Profile Editor.

12. BACKEND ARCHITECTURE

12.1 Server-Side Execution Flow

- **Request Ingestion:** Requests arrive at the Node.js server via HTTP/2.
- **SSR Page Rendering:** TanStack Start initializes the router, parses dynamic params (`{id, facultyId}`), fetches required server functions, executes DB queries in parallel, and returns hydrated HTML.
- **RPC Handling:** When client invokes a `createServerFn`, TanStack automatically sends a serialized POST request to `/_server/` endpoint, validating inputs via Zod before DB execution.
- **Connection Pooling:** Postgres.js client maintains an active connection pool with automatic health checks and parameterized queries preventing SQL injection.

13. FRONTEND ARCHITECTURE

13.1 Component Hierarchy

```
├── root.tsx (Root HTML Document, Global Styles, Head Metadata, Toaster)
│   ├── HeaderBanner (University Title, Circular Crest, Emergency Contacts)
│   ├── NoticeTicker (Live Marquee for Urgent Circulars)
│   ├── MegaMenu (Desktop Cascading Navigation / Mobile Responsive Drawer)
│   ├── PageProgressBar (Top-loaded navigation loading bar)
│   └── [ Dynamic Route Component (e.g., Department Home / Hostels / Placements) ]
│       ├── PageHero (Standardized Header Banner with Breadcrumbs)
│       ├── LocalSubNav / VerticalSubNav (Section-specific tab switcher)
│       └── BlockRenderer (Dynamic CMS JSON block builder)
│   ├── Chatbot (Persistent Floating JNTU AI RAG Assistant)
│   ├── AdminEditPanel (Floating quick-edit widget for logged-in admins)
│   └── Footer (University Address, Map, Accreditation, Quick Links, Credits)
```

13.2 Design System & Aesthetics

- **Color Tokens:** Primary Navy (`#0f2b48`), Accent Gold (`#d97706`), Surface Translucent Glass (`rgba(255,255,255,0.85)`), Dark Slate Ink (`#0f172a`).
- **Glassmorphism:** Micro-blur backdrops (`backdrop-blur-md`) with crisp sub-pixel border highlights (`1px solid rgba(255,255,255,0.2)`).
- **Responsive Viewports:** Tested across Mobile (375px+), Tablet (768px+), Desktop (1280px+), and 4K Displays.

14. AUTHENTICATION & ROLE-BASED ACCESS CONTROL (RBAC)

14.1 RBAC Matrix

Role	Identity Table	Login URL	Access Permissions
Super Admin	<code>admins</code>	<code>/admin</code>	Unrestricted global access: Notices, Tickers, Leadership, Gallery, RAG Re-indexing, System Logs.
Department HOD	<code>departments</code>	<code>/hod-login</code>	Scoped to assigned department slug: HOD message, Labs, Achievements, Gallery, Sidebar Nav, Block Pages.
Faculty Member	<code>faculty</code>	<code>/faculty-login</code>	Scoped to individual profile: Bio, Publications, Qualifications, Contact, Profile Avatar.
Public Visitor	None	Public Routes	Read-only access to all published institutional content.

14.2 Session Security Implementation

- **Password Hashes:** Hashed using `bcryptjs` with 10 salt rounds.

- **Session Tokens:** 256-bit cryptographically secure strings stored in `admin_sessions`.
- **Cookie Security:** Flags `HttpOnly; Secure; SameSite=Lax; Path=/; Max-Age=604800` (7 days).

15. ADMIN & CMS ARCHITECTURE

15.1 Super Admin Dashboard (`/admin`)

- **Notices Management:** Post, categorize, upload PDF attachments, and archive university circulars.
- **Ticker Management:** Add urgent scrolling banners with hyperlink targets.
- **Campus Gallery:** Upload and curate high-resolution photographs into categorized albums.
- **RAG Vector Synchronizer:** One-click execution of `ingest.server.ts` to re-embed modified circulars into `rag_chunks`.

15.2 Department Visual Page Builder

- HODs construct custom pages using structured UI blocks:
- `HeroBlock` : Full-width banner with background image and call-to-action.
- `RichTextBlock` : WYSIWYG formatted text with headings and lists.
- `FacultyGridBlock` : Auto-rendered grid of department faculty cards.
- `LabShowcaseBlock` : Grid of laboratory facilities with equipment lists.
- `DocumentTableBlock` : Downloadable list of PDF circulars and syllabus files.
- `GalleryGridBlock` : High-resolution photo masonry layout.
- `AccordionBlock` : Collapsible FAQ or syllabus modules.

16. FILE & MEDIA MANAGEMENT

16.1 Storage Directories

- **Static Assets:** Stored in `/public` (Logos, favicons, Lottie animations).
- **Uploaded Media:** Processed via `/api.upload.ts` and stored in `/local-assets/uploads/`.

16.2 Upload Constraints & Validation

- **Allowed MIME Types:** `image/jpeg`, `image/png`, `image/webp`, `image/svg+xml`, `application/pdf`.
- **Maximum File Size:** 15 Megabytes per upload.
- **Optimization:** Automated conversion and compression via `vite-plugin-imagemin` for production builds.

17. SECURITY ARCHITECTURE & THREAT MITIGATION

Threat Vector	Mitigation Strategy & Implementation
SQL Injection	100% Parameterized queries via Drizzle ORM and Postgres.js tagged template literals. No raw string interpolation.
Cross-Site Scripting (XSS)	React automatic output encoding + sanitization of rich-text HTML via DOMPurify before rendering.
Cross-Site Request Forgery (CSRF)	SameSite Lax / Strict cookies combined with custom header validation on all POST RPCs.
Brute Force Attacks	Bcrypt work factor 10 + automatic IP audit logging in <code>admin_audit_logs</code> .
Data Leakage	All credentials, database connection strings, and secret keys reside strictly in server environment variables. Zero credentials in client bundles.

18. DEPLOYMENT ARCHITECTURE

18.1 Deployment Options

Option A: Vercel Cloud Serverless Deployment (Configured)

- Commit code to GitHub repository (`Likhith32/JNTU-REDESIGN`).
- Link repository in Vercel Dashboard.
- Inject Environment Variables (`DATABASE_URL` , `ADMIN_SECRET` , `NODE_ENV`).
- Vercel utilizes `vercel.json` to build SSR output via `@cloudflare/vite-plugin` / TanStack Nitro.

Option B: Self-Hosted Linux VPS (Ubuntu 22.04 LTS / Node.js + PM2 + Nginx)

```
# Build the production bundle
npm run build

# Start process using PM2 process manager
pm2 start .vercel/output/functions/___server.func/index.mjs --name "jntugv-web"
pm2 save
pm2 startup
```

18.2 Nginx Reverse Proxy Configuration

```
server {
    listen 80;
    server_name jntugvcev.edu.in www.jntugvcev.edu.in;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name jntugvcev.edu.in www.jntugvcev.edu.in;

    ssl_certificate /etc/letsencrypt/live/jntugvcev.edu.in/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/jntugvcev.edu.in/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:8081;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache_bypass $http_upgrade;
    }
}
```

19. BACKUP & DISASTER RECOVERY

19.1 Automated PostgreSQL Database Backup

```
# Full compressed PostgreSQL backup command
pg_dump -U postgres -h localhost -d jntugv_db -F c -b -v -f /var/backups/jntugv_db_$(date +%Y%m%d_%H%M%S).dump
p

# Restoration command in disaster scenario
pg_restore -U postgres -h localhost -d jntugv_db -v -c /var/backups/jntugv_db_YYYYMMDD_HHMMSS.dump
```

19.2 Backup Schedule Recommendation

- Database Dump:** Daily at 02:00 AM (Retain last 30 daily backups).

- **Uploaded Media** (`/local-assets/uploads/`): Weekly incremental rsync backup to secondary storage server.
- **Git Source Code**: Mirrored to institutional GitHub/GitLab repository.

20. LOGGING & MONITORING

- **Application Logs**: Console output captured via PM2 / Vercel Runtime Logs.
- **Security Audit Logs**: Stored persistently in PostgreSQL `admin_audit_logs` table (Tracks IP, user agent, timestamp, action type).
- **Database Metrics**: Monitored via PostgreSQL active connection counts and query execution times.

21. TESTING & QUALITY ASSURANCE

- **Static Type Checking**: `npx tsc --noEmit` validates all TypeScript types across routes and components.
- **Linting & Code Style**: `npm run lint` executes ESLint rules and Prettier formatting checks.
- **Responsive Viewport Testing**: Validated across Chrome DevTools responsive presets (Mobile, Tablet, Desktop).
- **Cross-Browser Verification**: Tested on Google Chrome 120+, Microsoft Edge 120+, Mozilla Firefox 122+, Apple Safari 17+.

22. KNOWN ISSUES, TECHNICAL DEBT & LIMITATIONS

- **pgvector Cold-Start Latency**: Initial vector model download (`@xenova/transformers`) requires ~25MB ONNX model caching on server first run. Subsequent queries execute in <20ms.
- **Dynamic Upload Directory in Ephemeral Serverless**: When hosting on stateless serverless platforms (like Vercel), uploaded media in `/local-assets/uploads` should be mapped to an S3-compatible bucket (e.g., Cloudflare R2 / AWS S3) for permanent object retention.
- **Browser PDF In-Line Previews**: Mobile Safari forces download of large PDF circulars instead of inline rendering.

23. MAINTENANCE & DEVELOPER OPERATIONS MANUAL

23.1 How to Add a New Academic Department

- Insert new row in `departments` table via `src/db/schema.ts` or `psql ( slug , name , code , intake )`.
- Generate initial HOD password hash using Bcrypt and save in `hod_password`.
- Add department slug to MegaMenu and Department Index lists in `src/components/MegaMenu.tsx`.

23.2 How to Re-Index AI Chatbot Knowledge Base

```
# Run the local RAG ingestion worker
npx tsx scratch/run_ingest.ts
# This reads all notices, regulations, and departmental data, computes 384-dim embeddings, and updates rag_chunks.
```

23.3 How to Add a New CMS Block Type

- Define TypeScript block interface in `src/funcs/department-cms.server.ts`.
- Add block renderer component in `src/components/cms/BlockRenderer.tsx`.
- Register block editor controls in `src/components/cms/VisualPageBuilder.tsx`.

24. DEVELOPER TROUBLESHOOTING MATRIX

Problem / Symptom	Probable Cause	Corrective Action
500 Internal Server Error on API call	Database connection string missing or database unreachable	Verify DATABASE_URL in .env and verify PostgreSQL service is running (<code>sudo systemctl status postgresql</code>).
ERROR: type "vector" does not exist	pgvector extension not activated in database	Execute <code>CREATE EXTENSION IF NOT EXISTS vector;</code> in PostgreSQL database.
Chatbot returns outdated or empty responses	New circulars not yet ingested into vector table	Run <code>npx tsx scratch/run_ingest.ts</code> to refresh rag_chunks embeddings.
HOD Login fails with "Not Authenticated"	Missing or expired hod_session_dept cookie	Clear browser cookies, re-enter department credentials at /hod-login .
Image upload fails with 413 Payload Too Large	Upload exceeds 15MB limit or Nginx proxy limit	Adjust <code>client_max_body_size 20M;</code> in nginx.conf .

25. TECHNICAL HANDOVER SIGN-OFF CHECKLIST

- ☒ Complete source code delivered and verified on `main` branch.
- ☒ All 99 PostgreSQL database tables mapped, seeded, and verified.
- ☒ Vector embedding pipeline (`pgvector` + `@xenova/transformers`) active and tested.
- ☒ Super Admin (`/admin`), HOD (`/hod-login`), and Faculty (`/faculty-login`) portals operational.
- ☒ All 85+ public and dynamic department routes verified without broken links.
- ☒ Security policies (Bcrypt password hashing, SQL injection prevention, RBAC) validated.
- ☒ File upload pipeline with MIME type verification tested.
- ☒ Backup and restore scripts tested and documented.
- ☒ Strictly zero financial data included in documentation.

Handover Completed & Certified by Developers:

- Likhith Kumar Mankala
- Ch Sai Rupini
- P Anitha
- P Sai Vamsi

Project Guidance & Academic Supervision:

- Sri / Dr. W. Anil Sir (Faculty Coordinator / Assistant Professor)

Institution: Jawaharlal Nehru Technological University Gurajada Vizianagaram — College of Engineering Vizianagaram

Website: <https://jntugvcev.edu.in>